

Statement of Work

Background

The United States Department of Agriculture (USDA), Northeast Area (NEA) Dairy and Functional Foods Research Unit (DFFRU) at the Eastern Regional Research Center (ERRC) in Wyndmoor, PA has a need for a “Performance Maintenance with Full Repair Service Plan” for our UPLC/MS/MS instrument (Xevo TQ-S micro). The service plan must guarantee prompt repair/maintenance service with high quality when any repair is needed.

DFFRU scientists perform complex studies to identify and quantify the metabolites using this instrument while understanding how diet impacts the gut microbiota. Each study produces hundreds of samples that require comparing how the bacterial and fungal communities change over the course of a change in diet or treatment. This instrument requires very small amount of samples and provides the DFFRU scientists with timely and accurate analysis data within a very short period of time.

Interests

Scientists from the ERRC are interested in evaluation of the structural change in the gut microbial ecosystem in response to food, nutrition, and environmental stimuli, using next generation DNA sequencing technology and bioinformatics analysis.

What The Contractor will do

At the time of PM service, a trained engineer will replace all parts subject to wear as defined in the product specific PM checklist. All parts must be guaranteed genuine Waters quality parts and are included in a comprehensive PM parts kit. The requirements also include Calibration (Cal) which consists of a single point test which serves as a check of instrument performance, Qualification (Qual) which includes more comprehensive component level testing (Operational Qualification or OQ), as well as system level testing (Performance Qualification or PQ), plus Instrument and software upgrade. The requirements also include full repair service for the whole year on any additional service required during the contract period.

The vendor must use Waters’ appliances for the repairment or maintenance of this LCMS system and guarantee the quality. All key appliances of Waters’ instrument assembled are made by Waters and available only from Waters. Additionally, the service plan must guarantee high quality, prompt repair/maintenance service within 24 to 48 hours after notification.

The service Performance Maintenance Plan (PMP) and one PM is needed on the following items:

XEVO TQ-S Micro
UPLC H-CLASS Sample Manager
UPLC H-CLASS QSM
UPLC PDA Detector

What DFFRU will do

We will prepare samples of metabolites like- SCFAs, amino acids and bile acids profiles to identify and quantify them using the UPLCMS/MS instrument and to study the diet impacts on the gut microbiota from the data generated.